

EXPERIMENT – 5

Performing Various Configurations of IPTables Firewall in Kali Linux

Aim

To perform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Firewall Tool: IPTables

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

Introduction

IPTables is a powerful firewall utility in Linux used to configure rules for filtering incoming and outgoing network traffic. It operates by defining rules based on IP addresses, ports, and protocols. Configuring IPTables helps protect systems from unauthorized access, network attacks, and malicious traffic. Kali Linux provides built-in support for IPTables, making it suitable for learning firewall configurations in a controlled environment.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal and Check IPTables

1. Open **Terminal** in Kali Linux.
2. Check IPTables version:
3. `iptables --version`

Step 3: View Existing Firewall Rules

1. Display current IPTables rules:
2. `sudo iptables -L`

Step 4: Set Default Policies

1. Set default policy to drop incoming traffic:

2. `sudo iptables -P INPUT DROP`
3. Allow outgoing traffic:
4. `sudo iptables -P OUTPUT ACCEPT`

Step 5: Allow Localhost Traffic

1. Allow loopback interface traffic:
2. `sudo iptables -A INPUT -i lo -j ACCEPT`

Step 6: Allow SSH Access

1. Allow SSH connections on port 22:
2. `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

Step 7: Allow HTTP and HTTPS Traffic

1. Allow HTTP traffic:
2. `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`
3. Allow HTTPS traffic:
4. `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

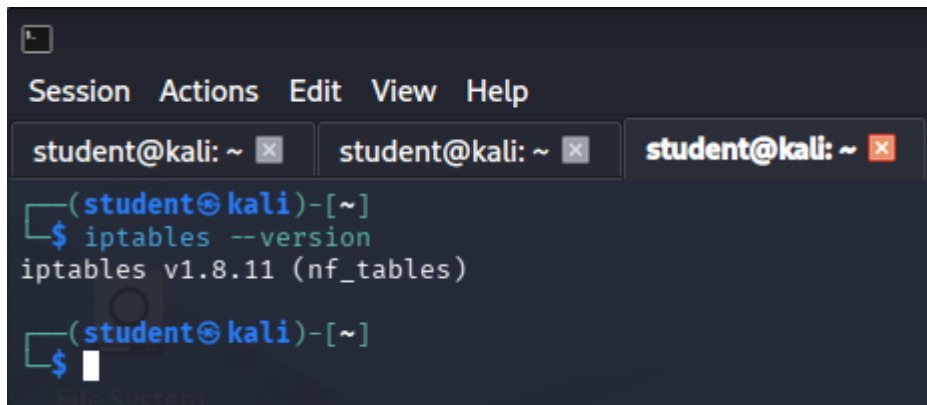
Step 8: Block a Specific IP Address

1. Block traffic from a specific IP:
2. `sudo iptables -A INPUT -s 192.168.1.100 -j DROP`

Step 9: Save IPTables Rules

1. Save the configured rules:
2. `sudo iptables-save > firewall_rules.txt`

Output

A screenshot of a terminal window with a dark background. At the top, there is a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. Below the menu bar, there are three tabs, each showing 'student@kali: ~' with a close button. The terminal content shows a prompt '(student@kali)-[~]' followed by the command '\$ iptables --version'. The output of the command is 'iptables v1.8.11 (nf_tables)'. Below this, another prompt '(student@kali)-[~]' is shown with a cursor, and the command '\$' is partially visible.

```
(student@kali)-[~]  
$ iptables --version  
iptables v1.8.11 (nf_tables)  
  
(student@kali)-[~]  
$
```

```

(student@kali)-[~]
$ sudo iptables -L
[sudo] password for student:

Sorry, try again.
[sudo] password for student:
reSorry, try again.
[sudo] password for student:
rec
sudo: 3 incorrect password attempts

(student@kali)-[~]
$ rec
rec:      SoX_ng v14.7.1.2

Usage summary: [gopts] [[fopts] infile]... [fopts] outfile [effect [effopt]] ...

SPECIAL FILENAMES (infile, outfile):
- Pipe/redirect input/output (stdin/stdout); may need -t
-d, --default-device Use the default audio device (where available)
-n, --null           Use the 'null' file handler; e.g. with synth effect
-p, --sox-pipe       Alias for '-t sox -'

SPECIAL FILENAMES (infile only):
"lprogram [options] ..." Pipe input from external program (where supported)
http://server/file         Use the given URL as input file (where supported)

GLOBAL OPTIONS (gopts) (can be specified at any point before the first effect):
--buffer BYTES           Set the size of all processing buffers (default 8192)
--clobber                Don't prompt to overwrite output file (default)
--combine concatenate    Concatenate all input files (default for sox, rec)
--combine sequence       Sequence all input files (default for play)
-D, --no-dither          Don't dither automatically
--dft-min NUM            Minimum size (log2) for DFT processing (default 10)
--effects-file FILENAME  File containing effects and options
-G, --guard              Use temporary files to guard against clipping
-h, --help               Display version number and usage information
--help-effect NAME       Show usage of effect NAME, or NAME=all for all
--help-format NAME       Show info on format NAME, or NAME=all for all
--i, --info              Behave as soxi(1)
--input-buffer BYTES     Override the input buffer size (default: as --buffer)
--no-clobber             Prompt to overwrite output file
-m, --combine mix        Mix multiple input files (instead of concatenating)
--combine mix-power      Mix to equal power (instead of concatenating)
-M, --combine merge      Merge multiple input files (instead of concatenating)
--magic                  Use 'magic' file-type detection
--multi-threaded         Enable parallel effects channels processing
--norm                   Guard (see --guard) and normalize

```

```

(student@kali)-[~]
$ sudo iptables -P INPUT DROP
[sudo] password for student:
rec
Sorry, try again.
[sudo] password for student:

```

Firewall rules are successfully applied.

Network traffic is filtered based on configured IPTables rules.

Result

Various firewall configurations were successfully implemented using IPTables in Kali Linux.

The experiment demonstrates how firewall rules can be used to control network traffic and improve system security.

Reference

<https://www.kali.org/docs/networking/iptables/>

Note

This experiment is performed **only for academic and ethical purposes** in a controlled laboratory environment.